

AVALON, Australia

WMO: 948540

Lat: 38.0286S

Lon: 144.4783E

Elev: 36

StdP: 14.68

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.5	37.1	31.9	26.3	46.8	33.9	28.6	46.7	31.8	53.5	29.2	54.1	4.6	300	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.5	96.0	67.2	90.0	65.9	84.5	64.8	70.8	83.7	69.0	81.8	67.4	79.5	17.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	101.8	73.5	65.4	93.8	72.2	63.8	88.8	71.0	34.7	83.6	33.1	81.8	31.8	79.7	92.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.8	25.6	23.0	DB	29.7	106.9	1.8	4.5	28.4	110.2	27.3	112.8	26.3	115.4	25.0	118.7	(4)
(5)			WB	29.3	77.4	1.7	6.1	28.1	81.8	27.1	85.4	26.1	88.8	24.9	93.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	67.8	67.9	64.9	59.1	54.4	50.2	49.5	50.4	53.4	57.0	60.9	64.3		
	DBStd	8.50	7.00	6.20	6.07	5.08	4.42	3.86	3.61	3.95	4.73	5.76	6.35	6.69		
	HDD50	164	0	0	0	1	9	43	54	41	13	3	0	0		
	HDD65	2914	41	33	77	191	331	443	481	452	349	262	159	95		
	CDD50	3181	552	502	460	274	144	51	38	54	116	218	328	444		
	CDD65	455	128	115	73	14	1	0	0	0	2	12	37	73		
	CDH74	4874	1268	988	711	179	5	0	0	1	36	293	496	897		
	CDH80	2275	672	453	307	49	0	0	0	0	6	105	211	472		
(14) Wind	WSAvg	11.5	12.6	12.1	11.4	10.1	10.5	9.8	11.2	11.9	11.9	11.9	11.8	12.3		
(15) Precipitation	PrecAvg	20.8	1.5	1.5	1.2	1.7	1.5	2.0	1.9	1.9	2.0	2.0	2.1	1.5		
	PrecMax	28.3	4.6	5.1	3.3	6.4	3.0	4.2	3.5	3.1	3.9	4.6	4.4	4.4		
	PrecMin	12.3	0.1	0.1	0.4	0.4	0.4	0.9	0.7	0.9	0.9	0.3	0.6	0.1		
	PrecStd	4.0	1.1	1.1	0.7	1.2	0.8	0.8	0.8	0.6	0.8	1.2	1.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.8	98.5	96.9	85.9	73.8	64.8	64.4	69.9	78.6	89.9	93.0	100.9		
		MCWB	69.0	67.4	67.6	62.3	58.0	55.0	53.2	56.6	59.0	61.7	65.2	66.7		
	2%	DB	95.7	91.8	87.9	79.0	69.4	61.2	60.9	64.6	71.9	81.9	86.4	92.8		
		MCWB	67.8	66.9	65.7	60.1	57.2	54.0	52.5	53.7	57.0	59.8	65.0	66.2		
	5%	DB	86.5	85.7	81.7	73.5	65.9	59.2	58.7	61.2	67.6	75.3	79.2	84.0		
		MCWB	67.0	66.4	64.3	59.9	56.2	53.1	51.3	52.4	56.0	59.0	62.7	64.9		
	10%	DB	79.2	79.1	75.6	69.4	62.7	57.4	56.8	58.8	63.7	69.7	73.1	76.6		
		MCWB	66.1	66.1	63.5	58.9	55.4	52.1	50.5	51.3	54.3	57.4	61.5	63.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.9	72.0	70.3	65.2	61.1	57.3	55.8	57.7	61.1	65.4	70.0	72.8		
		MCDB	86.5	83.6	84.0	78.3	67.6	61.5	60.9	66.4	72.5	78.3	83.8	86.7		
	2%	WB	71.1	70.0	67.9	63.0	59.0	55.5	53.7	55.1	58.8	62.2	66.8	68.6		
		MCDB	84.1	82.0	80.1	71.5	66.3	59.4	58.6	62.3	68.4	75.2	79.3	83.2		
	5%	WB	68.9	68.5	66.3	61.3	57.3	54.1	52.2	53.2	56.7	60.1	64.5	66.4		
		MCDB	82.4	79.7	77.0	69.7	63.7	58.3	57.0	59.7	65.5	72.4	75.7	79.3		
	10%	WB	66.8	67.0	64.7	59.8	55.9	52.8	51.1	51.8	54.9	58.1	62.3	64.4		
		MCDB	79.1	77.5	73.8	67.7	61.5	56.9	55.7	57.6	62.4	68.3	71.3	75.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.4	19.5	19.2	18.4	15.8	14.3	14.6	16.1	18.5	21.1	20.1	21.1		
		MCDBR	35.9	32.4	31.1	27.5	20.7	16.0	16.8	20.5	25.8	33.9	32.8	36.6		
		MCWBR	12.7	11.5	12.5	12.6	11.6	10.5	10.2	12.1	13.6	15.4	14.0	13.8		
	5% WB	MCDBR	28.1	24.0	24.4	21.8	17.6	15.2	15.1	18.4	22.6	27.9	26.9	27.7		
		MCWBR	11.8	10.5	10.7	11.4	10.8	10.7	10.1	11.6	12.6	14.3	12.8	12.6		
(40) Clear-Sky Solar Irradiance	taub		0.353	0.356	0.334	0.328	0.315	0.302	0.303	0.313	0.334	0.340	0.349	0.345		
	taud		2.497	2.498	2.559	2.549	2.566	2.591	2.580	2.543	2.475	2.474	2.478	2.510		
	Ebn at Noon		311	302	296	278	263	258	264	279	291	304	310	315		
	Edh at Noon		36	34	30	26	23	20	22	26	32	35	36	36		
(44) All-Sky Solar Radiation	RadAvg		2257	1974	1559	1086	728	569	638	898	1284	1708	1989	2238		
	RadStd		139	93	89	59	37	41	36	52	82	126	117	128		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	+0.90	N/A	N/A	+1.58	N/A	N/A	-31	-232	+315	+106

Nomenclature: See separate page